EXCEL 2

HOANG THI KIEU HOA

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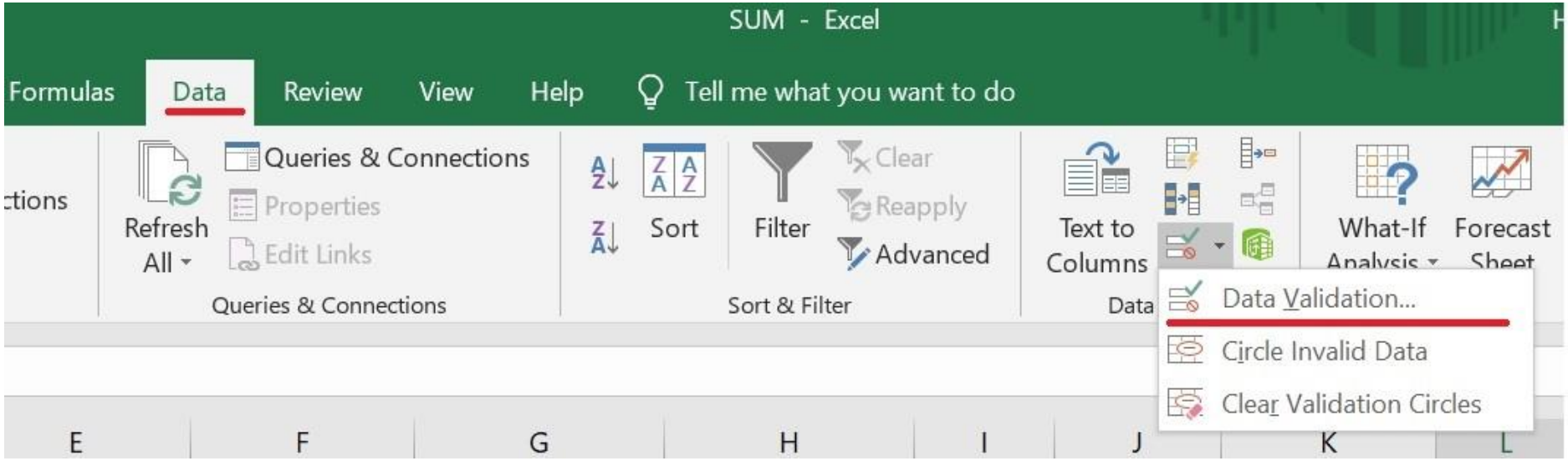
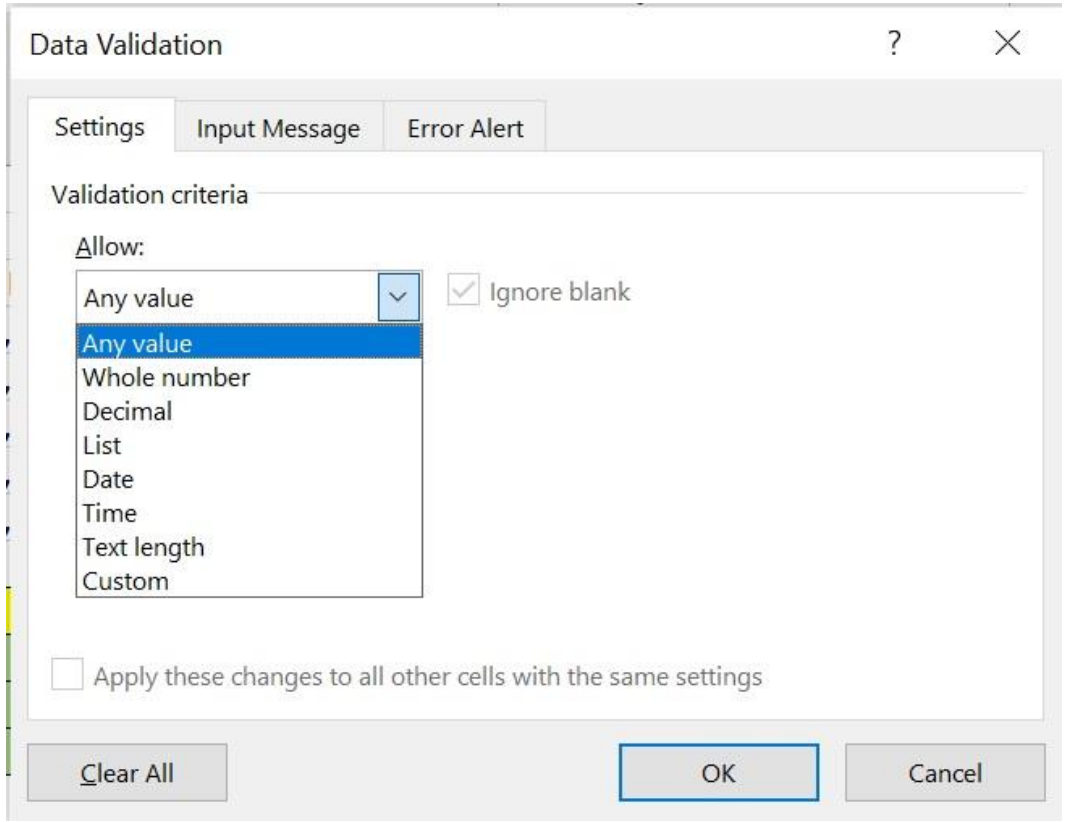
- Create Chart
- Lookup functions

Data Validation

Data Validation

- Data Validation in Excel lets you control the data that can be entered in a cell. You can restrict the user to enter only a specified range of numbers or text or date.
- When you want to restrict data entry to certain numbers/text/dates. Data that does not meet the validation criteria is not allowed.
- When you want to inform user whenever out-of-range data is entered. However, all kinds of data entry is allowed.
- When you guide the user on what data to enter. All kinds of data entries are allowed.


SUM - Excel
Formulas
Data
Review
View
Help
Tell me what you want to do
Refresh All
Queries & Connections
Properties
Edit Links
Queries & Connections
Sort
Filter
Clear
Reapply
Advanced
Sort & Filter
Text to Columns
Data
What-If Analysis
Forecast Sheet
Data Validation...
Circle Invalid Data
Clear Validation Circles
E
F
G
H
I
J
K
L

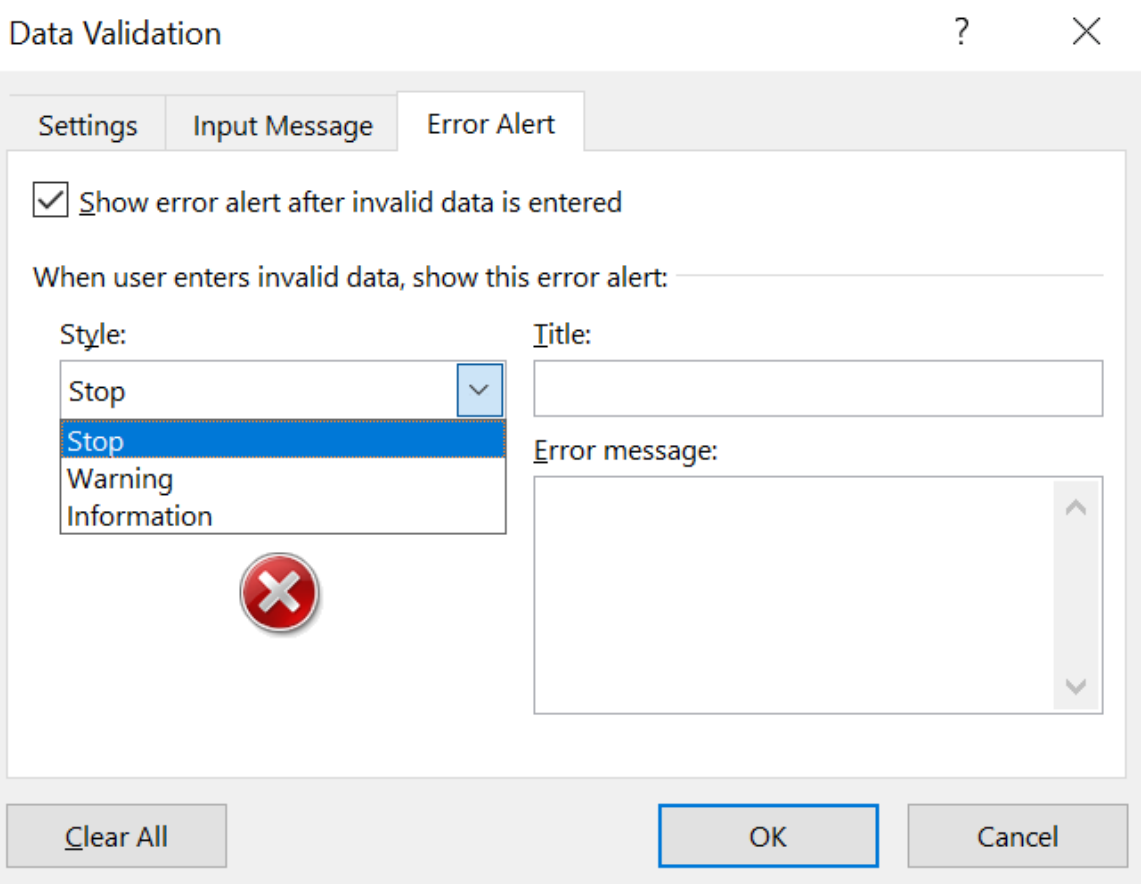

Data Validation
Settings
Input Message
Error Alert
Validation criteria
Allow:
Any value
Ignore blank
Any value
Whole number
Decimal
List
Date
Time
Text length
Custom
Apply these changes to all other cells with the same settings
Clear All
OK
Cancel

Data Validation

- In the drop-down list, you can choose the condition you want to apply for a range of cells
- Any Value – Allows any value to be entered in a cell.
- Whole Number – Allows only whole numbers to be entered, with additional conditions such as greater/less than, between/not between, equal to/not equal to.
- Decimal – Allows numbers with decimals to be entered, with additional conditions such as greater/less than, between/not between, equal to/not equal to.
- List – Creates a drop down by taking a list of items (through range selection or named range).
- Date – Allows dates (or its number value) to be entered, with additional conditions such as greater/less than, between/not between, equal to/not equal to.
- Time – Allows time (or its number value) to be entered, with additional conditions such as greater/less than, between/not between, equal to/not equal to.
- Text Length – Allows text with the condition on its length.
- Custom – Allows values that meets the specified criteria. For example, if I use the formula =A1>10, then only numbers greater than 10 are allowed in cell A1.

Data Validation

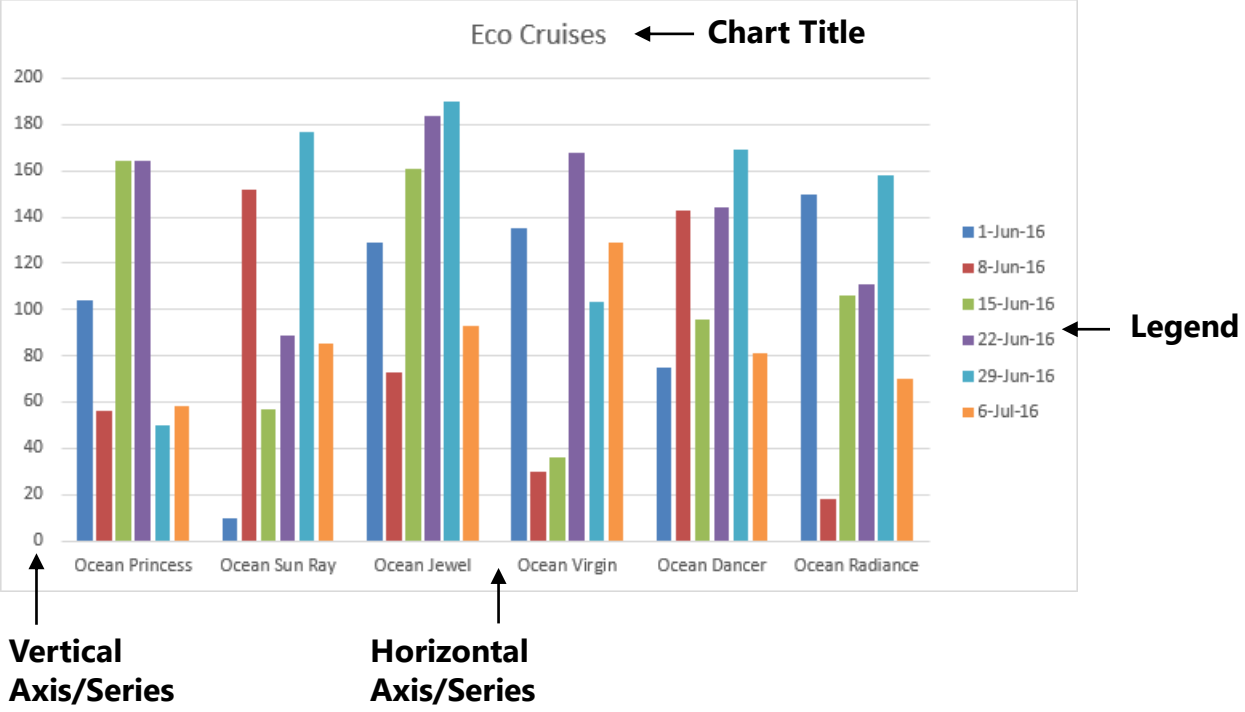
- You can customize the error message by going to the Error Alert tab in the Data Validation dialogue box. There are three options:
- **Stop Error** – Displays the stop error and does not let user enter the data which is out of the specified range.
- **Warning Error** – Displays the warning error but lets user enter the data which is out of the specified range.
- **Information Error** – Displays the information error but lets user enter the data which is out of the specified range.



WORKING WITH CHART

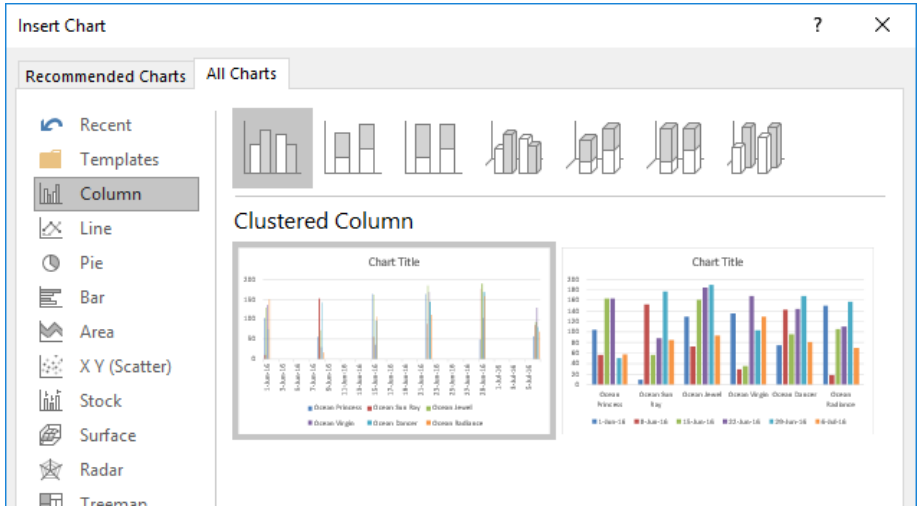
Working with Charts

- A chart can illustrate trends or patterns in the data.
- You can create a chart by selecting data from the worksheet, and then select the type of chart to insert



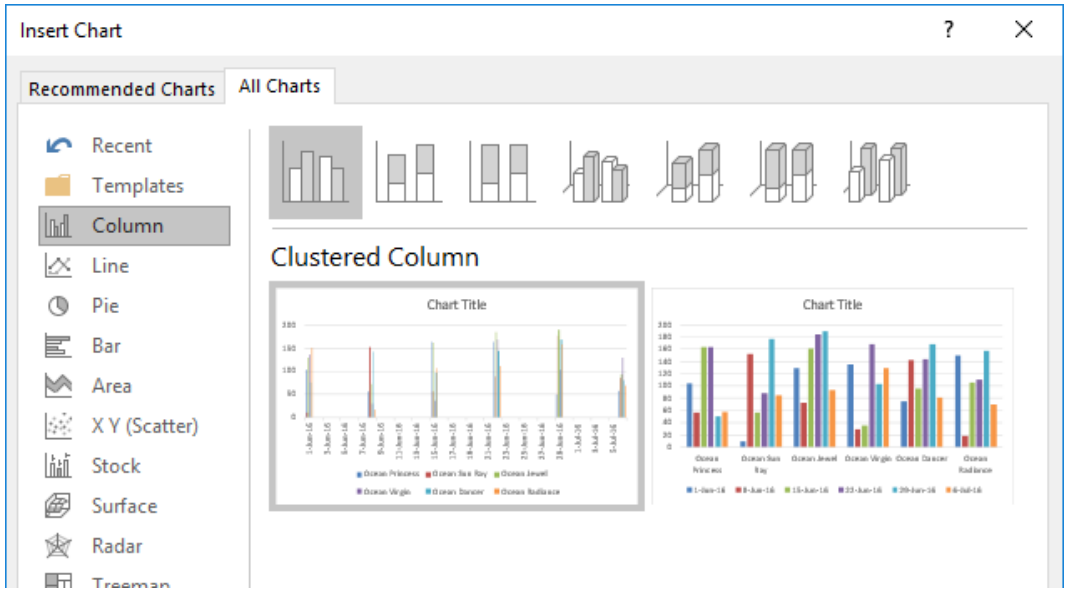
Working with Charts

- **Selecting Chart Types**
  - Line charts are better for trends, bar charts are better for volume, and pie charts are best for showing portions of a total
  - Excel provides a variety of chart types and several subtypes within each major type



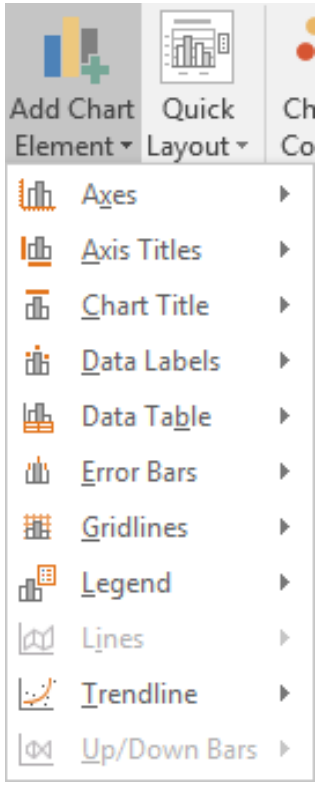
Working with Charts

- **Selecting Chart Types**
  - **Column** – compares values over time
  - **Line** – compares continuous trends
  - **Bar** – compares values over time (horizontal)
  - **Area** – compares a continuous change in volume
  - **X Y (Scatter)** – determines data patterns
  - **Stock** – displays high-low-close data
  - **Surface** – displays trends in 3-D
  - **Radar** – determines patterns with points and lines



Working with Charts

- **Changing the Chart Layout**
  - **Axes** – Include labels on the horizontal and vertical axes
  - **Axis Titles** – Add titles to the horizontal and vertical axes
  - **Chart Title** – Add a title for the chart
  - **Data Labels** – Include data labels on the chart
  - **Data Table** – Display the chart data beneath the chart
  - **Error Bars** – Displays margins of error and standard deviations
  - **Gridlines** – Include gridlines on the chart
  - **Legend** – Include a legend and position it in relation to the chart



VLOOKUP - HLOOKUP

VLOOKUP

- The VLOOKUP function in Excel is a tool for looking up a piece of information in a table or data set and extracting some corresponding data/information.
- For example: the VLOOKUP function says the following to Excel: “Look for this piece of information (e.g., bananas), in this data set (a table), and tell me some corresponding information about it (e.g., the price of bananas)”.

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

- **Lookup_value** (required argument) – Lookup_value specifies the value that we want to look up in the first column of a table.
- **Table_array** (required argument) – The table array is the data array that is to be searched. The VLOOKUP function searches in the left-most column of this array.
- **Col_index_num** (required argument) – This is an integer, specifying the column number of the supplied table_array, that you want to return a value from.
- **Range_lookup** (optional argument) – This defines what this function should return in the event that it does not find an exact match to the lookup_value. The argument can be set to TRUE or FALSE, which means:
  - TRUE – Approximate match, that is, if an exact match is not found, use the closest match below the lookup_value.
  - FALSE – Exact match, that is, if an exact match not found, then it will return an error.

=VLOOKUP(A11,A4:C8,3,FALSE)

	A	B	C	D	E	F	G	H
1	**Good Table**				**Bad Table**			
2								
3	**Fruit**	**In Stock?**	**Price ($/lb)**		**In Stock?**	**Price ($/lb)**	**Fruit**	
4	Grapes	Yes	7.25		Yes	7.25	Grapes	
5	Mangos	Yes	12.32		Yes	12.32	Mangos	
6	Bananas	Yes	5.42		Yes	5.42	Bananas	
7	Pineapple	No	7.18		No	7.18	Pineapple	
8	Oranges	Yes	8.99		Yes	8.99	Oranges	
9								
10	**Fruit**	**Price**			**Fruit**	**Price**		
11	Bananas	4:C8,3,FALSE)			Bananas	#N/A		
12								
13								
14								
15								

HLOOKUP

- **HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])**
- **Lookup_value** (required argument) – Lookup_value specifies the value that we want to look up in the first column of a table.
- **Table_array** (required argument) – The table array is the data array that is to be searched. The VLOOKUP function searches in the left-most column of this array.
- **Row_index_num** (required argument) – This is an integer, specifying the row number of the supplied table_array, that you want to return a value from.
- **Range_lookup** (optional argument) – This defines what this function should return in the event that it does not find an exact match to the lookup_value. The argument can be set to TRUE or FALSE, which means:
  - TRUE – Approximate match, that is, if an exact match is not found, use the closest match below the lookup_value.
  - FALSE – Exact match, that is, if an exact match not found, then it will return an error.

B8 =HLOOKUP("D",A1:F5,4,FALSE)

	A	B	C	D	E	F	G
1	**Student name**	**A**	**B**	**C**	**D**	**E**	
2	**Accounts**	75	65	70	60	59	
3	**Economics**	65	72	78	89	67	
4	**Management**	70	68	90	72	58	
5	**Mathematics**	80	90	75	65	87	
6							
7							
8	**Fetch Marks of D in Management**	72					
9							